

- All steps must be shown clearly.
- Formulae used must be written down clearly. Name the variables used.

Q1. In a city, chances of rain on any day in November is 10%.

i) Why is this a Binomial process? Thereby, what are the chances that November 2026 will experience 5 days of rain?

ii) Treat the same as a Poisson process.

iii) Is the Poisson approximation appropriate in this case? Comment.

(2+2+2 = 6 marks)

Ans. i) *Binomial process with 2 kinds of events:*

Rain
No Rain

1/2 mark

$$\text{Prob. of rain, } s = 10\% \\ = 0.1$$

$$\text{Prob. of no rain} = 0.9$$

$$\text{Total no. of days} = 30$$

$$\text{No. of days of rain, } n = 5$$

Using Binomial Distribution:

$$P(N; n, s) = {}^N C_n (s)^n (1-s)^{N-n}$$

$$= {}^{30} C_5 (0.1)^5 (0.9)^{25}$$

$$= (142506) \times (10^{-5}) \times (7.17897 \times 10^{-2})$$

$$= 10230.477 \times 10^{-5}$$

$$= 0.1023$$

1 mark

1/2 mark

$$\text{ii) Now, } N \cdot s = 30 \times 0.1 = 3 < 5$$

So, we may try the Poisson distribution
For a mean no. of events λ , the probability of

'n' successes is given as,
$$P(n, \lambda) = \frac{\lambda^n e^{-\lambda}}{n!}$$

Hence, $\lambda = 30 \times 0.1 = 3$ 1/2 mark
 $n = 5$

So, $P(5, 3) = \frac{3^5 e^{-3}}{5!} = (243) \times (0.04978706)$ 1 mark
 $= 0.100818$ 1/2 mark

iii) Yes/No
 $N \geq 100$
 $P \leq 0.05$

2 marks

The agreement betⁿ the Binomial and the Poisson process is good. The Poisson approximation holds.

Note: Marks were given for logical explanations for both the answers.

Q2. A scientist studies the growth rate of mangroves. He traced the increase in length, in cm, of five saplings over one year in a location in India.

His recorded data was {52.5, 48.8, 44.6, 55.5, 45.6}.

Globally, the mean height increase is known to be 53.0 cm in one year.

i) Record the sample statistics.

ii) State hypotheses to evaluate if this cohort has lower growth rate than the global reference. value.

iii) Evaluate your hypothesis with a suitable test. State why you chose this particular test.

iv) Schematically depict the relevant distribution, marking the "acceptability region".

(2+1+4+1 = 8 marks)

i) Sample statistics

$$n = 5$$

$$v = 4$$

$$\text{Mean, } \bar{x} = 49.4 \text{ cm}$$

$$\text{Variance, } s^2 = 21.165$$

1 mark

$$\text{Std. dev, } S = 4.60054$$

$$\text{Std. error, } SE = S/\sqrt{n} \\ = 2.057424$$

1 mark

ii) Reference level is $\mu_0 = 53.0 \text{ cm}$
Constructing the hypothesis

$$H_0: \mu = \mu_0$$

$$H_A: \mu < \mu_0$$

1 mark

iii) One tailed t-test, due to small sample size
C.I = 95%, $\alpha = 0.05$

$$t_{\text{score}} = \frac{\bar{x} - \mu_0}{SE}$$

1 mark

$$= \frac{49.4 - 53.0}{2.057424}$$

$$t_{\text{score}} = -1.749760 \approx -1.750$$

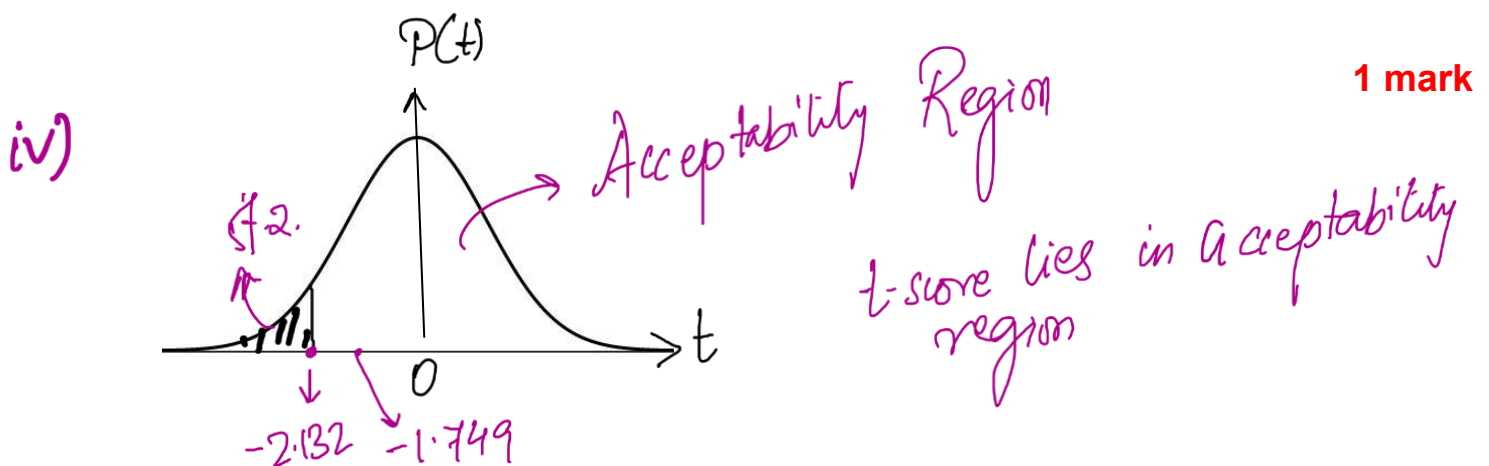
1 mark

We will compare $|t_{\text{score}}|$ to $t_{0.05,4}^{(1)} = 2.132$

1/2 mark

→ Hence, we retain H_0

1/2 mark



1 mark

Q3. A large sample of 100 cancer patients is randomly selected to find the weight change in 2 weeks from the intake of a food supplement. The mean weight change was +1.2 kg, and std. dev. was 0.8 kg. At the population level, the mean weight change of cancer patients is -0.5 kg (in the same time).

i) State hypotheses to evaluate if weight change of this sample is *different* from the reference.

ii) Evaluate your hypothesis at a confidence interval of 95%.

(2+6 = 8 marks)

i) Hypothesis

$$H_0 : \mu = \mu_0 (-0.5)$$

$$H_A : \mu \neq \mu_0 (-0.5)$$

2 marks

NOTE: Mentioning any value other than -0.5 leads to -1/2 mark deduction. Not mentioning any value is okay.

ii) Sample Statistics

$$n = 100 (>30)$$

$$\text{mean } \bar{x} = 1.2 \text{ kg}$$

$$\text{std. dev } s = 0.8 \text{ kg}$$

$$\text{var } s^2 = 0.64 \text{ kg}^2$$

$$S.E. = \frac{s}{\sqrt{n}} = \frac{0.8}{\sqrt{100}} = \frac{0.8}{10} = 0.08$$

2 marks

We wish to evaluate whether this sample is different from the population, reference value, $\mu_0 = -0.5 \text{ kg}$. Since the sample is large ($n > 30$), we will use the Z-test.

$$Z\text{-score} = \frac{\bar{x} - \mu_0}{SE} = \frac{1.2 - (-0.5)}{0.08} = 21.25$$

2 marks

The C.I. at 95%, i.e., $\alpha = 0.05$ for a 2-tailed Z-test

$$Z_{\alpha/2} = 1.96$$

1 mark

Now, $|21.25| > 1.96$. We reject H_0 or accept H_A

1 mark

Q4. A nutritionist is interested in comparing the protein content (arb. units) in two legume species. She tested two random samples, whose statistics are reported below:

	Legume-1	Legume-2
Sample size	8	8
Mean	18.2	14.4
Variance	4.0	3.2

She supposes that at the population level, there is no difference in their mean protein content.

i) State hypotheses for the nutritionist. State the most appropriate test in this case.

iii) Evaluate the hypothesis based on the data provided at $\alpha = 0.05$.

iv) Schematically depict the relevant distribution, marking the "acceptability region"

(2+5+1 = 8 marks)

Given,
 $n_1 = n_2 = 8$
 $\bar{x}_1 = 18.2, S_1^2 = 4.0$
 $\bar{x}_2 = 14.4, S_2^2 = 3.2$

ii) Hypothesis

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

Most Appropriate Test : Two Sample t-test with pooled variance 1 mark

cii) Pooled variance :
$$var = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

$$= \frac{7 \times 4.0 + 7 \times 3.2}{14} = 3.6$$
 1 mark

Standard Error $SE = S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$

$$= 1.897 \sqrt{0.28} = 0.9487$$
 1 mark

t score, $t = \frac{\bar{x}_1 - \bar{x}_2}{SE} = \frac{18.2 - 14.4}{0.9487} = 4.01$ 1 mark

Degrees of freedom, $\nu = n_1 + n_2 - 2 = 14$

1 mark

$$\alpha = 0.05$$

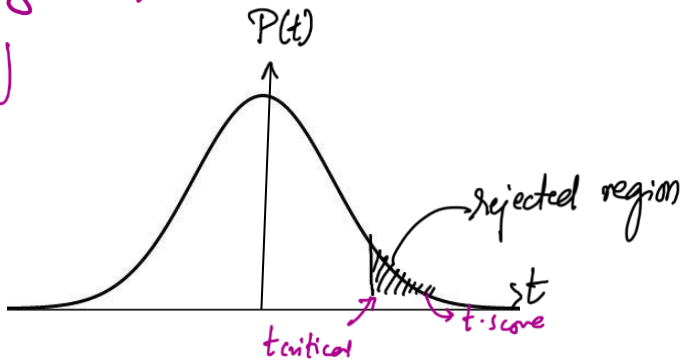
$$t_{critical} = t_{\alpha/2, \nu} = t_{0.025, 14} = 2.145$$

Since, $t_{score} > t_{critical}$, Reject H_0

1 mark

(iii)

1 mark



Formula Sheet

$$Z = \frac{\bar{X} - \mu}{\sigma_{\bar{X}}}$$

$$t_{score} = \frac{[\bar{X}_1 - \bar{X}_2] - [\mu_1 - \mu_2]}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$t_{score} = \frac{[\bar{X}_1 - \bar{X}_2] - [D_0]}{\sqrt{s^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$\text{approx.} \approx \frac{\left[\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right]^2}{\frac{\left(\frac{s_1^2}{n_1} \right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2} \right)^2}{n_2 - 1}}$$

$$\text{var} = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 + n_2 - 2)}$$

$$t_{score} = \frac{[\bar{X}_1 - \bar{X}_2] - [D_0]}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)}}$$